How can the composting process be made more effective?

堆肥化を効率化させる方法

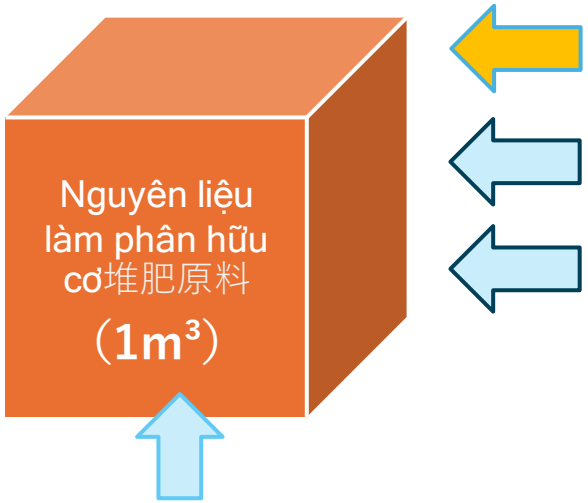
To make the composting process more effective,

(1) **Adequate nutrient sources:** Use fresh animal waste (organic matter from livestock manure) as much as possible.
(2) **Temperature:** 10°C or higher is ideal. If the temperature is too low, bacterial activity will decrease.
(3) **Appropriate moisture:** 55-72%. If it is too high, proper aeration cannot occur. If it is too low, bacterial activity will decrease.
(4) **Sufficient oxygen (air):** 50 to 100 L/min/m³. If it is too low, the environment becomes anaerobic and anaerobic decomposition occurs.

Lack of oxygen can cause microbial death, while too much oxygen will result in heat loss and wasted energy.

堆肥化を効率化させるためには、
(1) 十分な栄養源→できるだけ新鮮なふん(家畜ふんの有機物）利用（ウッドチップ、等）
(2) 温度→10°C以上が望ましい。低すぎると微生物活性が低下
(3)適当な水分→55～72%、高すぎると通気できない。低すぎると微生物活性が落ちる。
(4)十分な酸素（空気）→50～100L/min/m3、少なすぎると嫌気状態になり嫌気分解する、
酸素不足は致命的。多すぎると熱を奪う、無駄なエネルギーを消費。

Image



Activate aerobic microorganisms

好気性微生物の活性化

Provide enzymes　酵素の供給

ウッドチップ、等

Mix thoroughly.　切り返し作業

Ensure proper aeration (improve physical properties).
通気性の確保（物理性の改❹）

Air
空気50～100L/min/m³

Mix returned compost with secondary materials (to adjust moisture).
戻し堆肥や副資材の混合（水分調整）

※ Requirements for Returned Organic Fertilizers and Secondary Materials

※戻し堆肥や副資材の要件

〇安価で入手しやすいこと

※Examples of auxiliary materials

※副資材の例

Sawdust, rice husks, pruned branches, coffee grounds, wood chips, etc.

おがくず、もみ殻、剪定枝、コーヒー粕、ウッドチップ、等

○Low moisture (water content)
○水分（含水率）が低いこと

○High porosity
○空隙性が高いこと

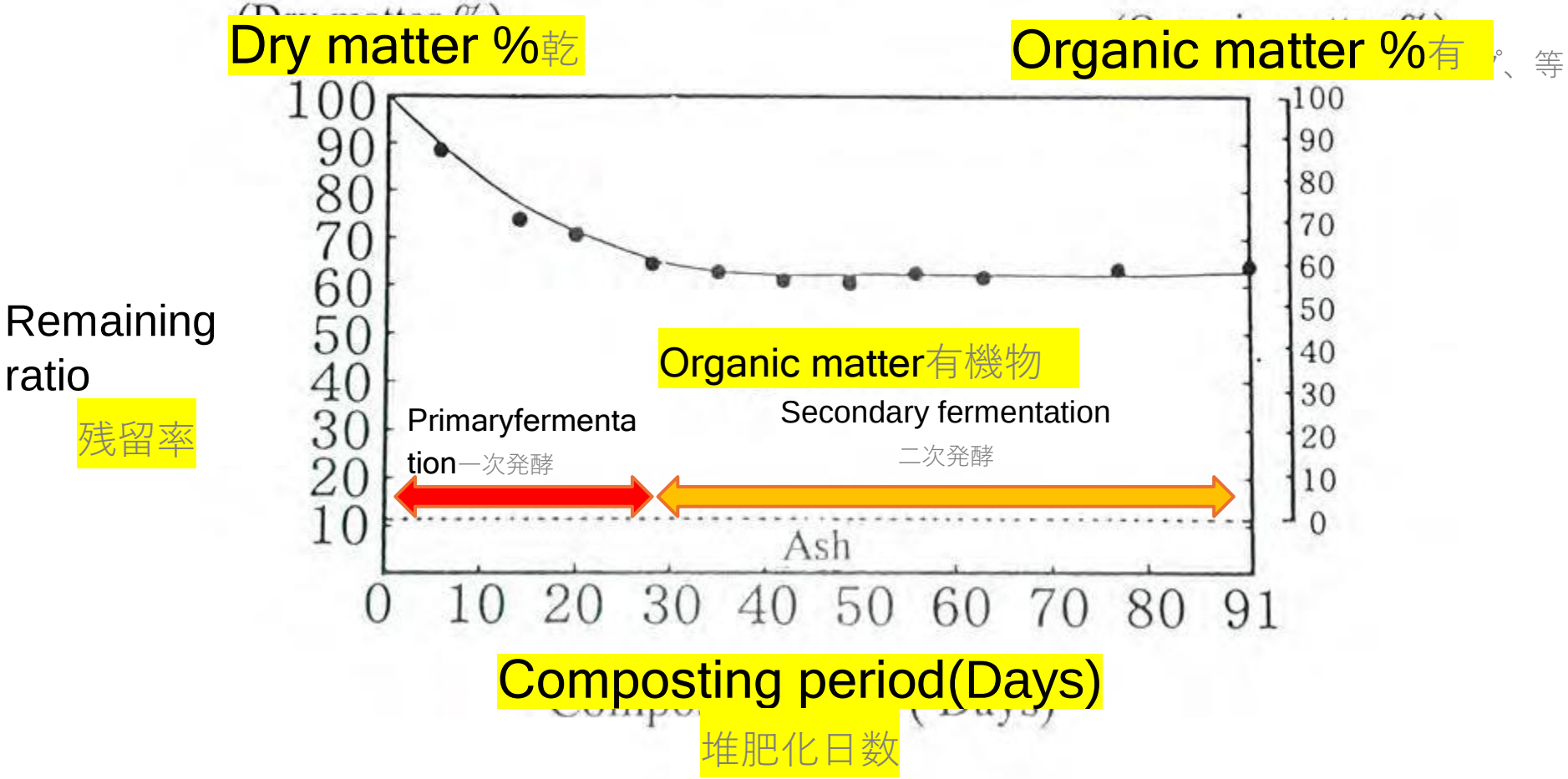
○This is a safe organic material.
○安全な有機物であること

○Cheap and easy to find.
○安価で入手しやすいこと

Thời gian quá trình ủ phân và phân hủy chất hữu cơ (ví dụ)

堆肥化過程の期間と有機物分解（例）

Pig manure + sawdust 豚糞＋おがくず



Usage Biso Enzyme 美創酵素の使用方法

When mixing the compost, dilute the concentrate with water 1,000 to 2,000 times and spray evenly using a fine-mesh sprayer.

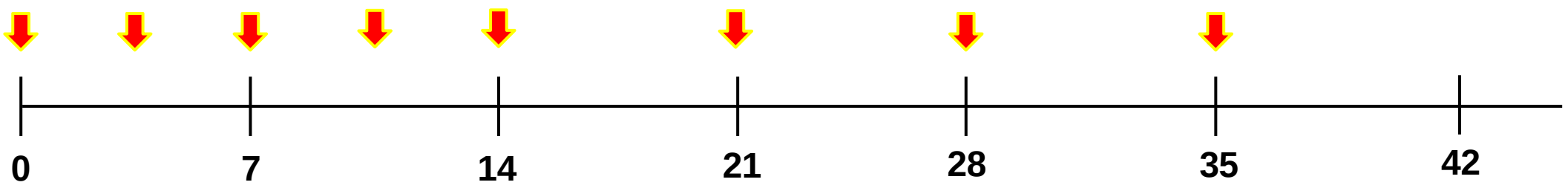
The recommended spraying volume (diluted product) is 2 to 4 liters per 1 m^3 (ton) of raw compost material.

ウッドチップ、等

(Stock solution volume: 2-4L ÷ 1,000-2,000 times = 1-4ml/m^3(t))

堆肥攪拌時に製品原液を水で1,000～2,000倍にうすめて、細かい目のスプレーでまんべんなく噴霧します。
噴霧水量（製品希釈水）の目安は、堆肥原料1m³(t)あたりに2～4Lです。
（製品原液量・・・2～4L÷1,000～2,000倍＝1～4ml/m³(t)）

When to use enzymes（**Encourage**）　酵素を使用するタイミング（推奨）



Composting period (Days)

堆肥化日数

※ However, please consider the following factors: compost moisture, odor intensity, weather, and temperature.

但し、たい肥の水分率などの状態、悪臭の強度、天候や気温などに応じて検討してください。

Reference) Various mixing methods and equipment

Ventilated compost shed and skid-steer loader
(4-wheel skid-steer system)

Mixing with a loader (or **Excavator mixing**)

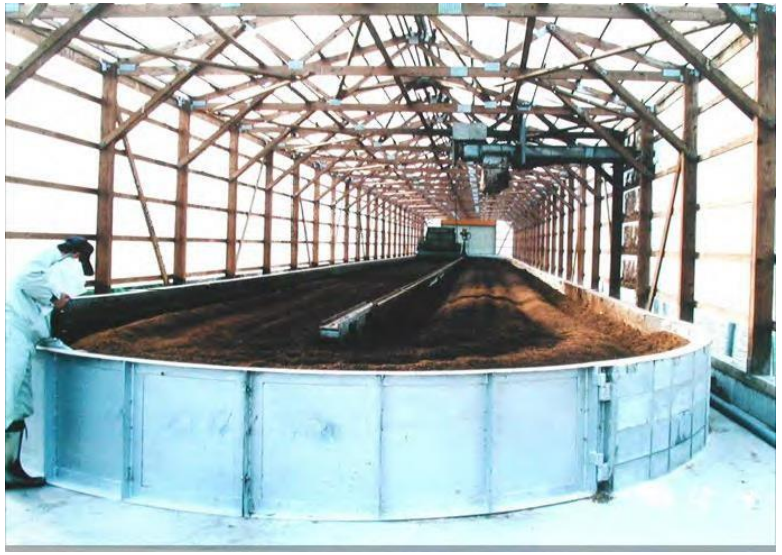
Mixing with a scoop-type mixer

Mixing with a self-propelled

Mixing with a rotary

Screw mixer

Reference) Various mixing methods and equipment



Rotary drum fermentation tank composting equipment (or **Rotary fermentation vessel**)

Vertical closed composting equipment (or **Closed vertical composting vessel**)

ウッドチップ、等

The contents of this document are basic information and are intended to serve as a reference or guide. Additionally, the method and amount of enzyme product used must be adjusted depending on the actual site conditions, the properties of the target materials, the budget, etc.

本資料の記載内容は、基本事項と参考や目安になる情報です。
実際の現場条件や対象物の性状、予算などに応じて、酵素製品の使用方法や使用量をアレンジする必要もあります。

ウッドチップ、等